

Entanglement as a Resource: Superdense Coding, Teleportation, and Quantum Key Distribution

Rajarshi Giri,^{*} Arkaprava Bose,[†] Rudra Pal,[‡] and Sarbajit Sarkar[§]

*Department of Physical Sciences, Indian Institute of Science
Education and Research Kolkata, Mohanpur 741246, India*

(Dated: April 30, 2026)

Quantum entanglement gives rise to correlations that cannot be explained within any classical framework and serves as a resource in (quantum) communication and cryptography. We review the conceptual foundations underlying entanglement and nonclassical features of entanglement through the EPR paradox, Bell's theorem, the CHSH inequality, and the CHSH game. We then study two landmark applications: superdense coding, which uses preshared entanglement to transmit two classical bits via a single qubit, and quantum teleportation, which enables the transfer of an unknown quantum state through entanglement and classical communication. Finally, we discuss the role of entanglement in secure communication through the BB84 and E91 quantum key distribution protocols. `Qiskit` simulations and `IBM Quantum` hardware implementations are presented to illustrate these protocols and their performance on real quantum devices.

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^{*} rg23ms099@iiserkol.ac.in

[†] ab23ms037@iiserkol.ac.in

[‡] rp23ms085@iiserkol.ac.in

[§] ss23ms097@iiserkol.ac.in;

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I. QUANTUM ENTANGLEMENT

One of the fundamental questions of modern physics is to determine what precisely differentiates the quantum world from the classical one. *Quantum entanglement* is a phenomenon that is not possible in our classical world. The departure from classical lines of thought is highlighted most emphatically by the property of entanglement.

A. The Mathematical framework of entanglement

Quantum entanglement is a property that belongs to two or more quantum states.

Hilbert space of combined state: Consider two quantum systems with associated Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$. The Hilbert space of the combined system is given by the tensor product of $\mathcal{H}_1$ and $\mathcal{H}_2$.

$$\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$$

Separable pure state: A pure state $|\Psi\rangle$ within the combined system $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ is defined as *separable pure state* if $|\Psi\rangle$ can be factored in the form $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, where $|\psi_1\rangle \in \mathcal{H}_1$ and $|\psi_2\rangle \in \mathcal{H}_2$.

Entangled pure state: A pure state $|\Psi\rangle$ within the combined system $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$ is defined as an *entangled pure state* if $|\Psi\rangle$ cannot be factored in the form $|\Psi\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$, where $|\psi_1\rangle \in \mathcal{H}_1$ and $|\psi_2\rangle \in \mathcal{H}_2$.

B. Examples of Quantum entanglement

Let's assume 2 particles, each of which comprises a qubit. States of particles 1 and 2 are within the Hilbert spaces $\mathcal{H}_1$ and $\mathcal{H}_2$ respectively.

$$|\psi_1\rangle \in \mathcal{H}_1 \quad \text{and} \quad |\psi_2\rangle \in \mathcal{H}_2$$

We take the basis of these Hilbert spaces to be the spin in the z -direction, with eigenstates spin-up $|\uparrow\rangle$ or spin-down $|\downarrow\rangle$.

The Hilbert space of this combined system is given by, $\mathcal{H} = \mathcal{H}_1 \otimes \mathcal{H}_2$

Assume, $|\psi_1\rangle = |\uparrow\rangle$ and $|\psi_2\rangle = |\downarrow\rangle$. In this case, the state of the combined system is $|\Psi\rangle = |\uparrow\rangle \otimes |\downarrow\rangle$. Thus, it is an example of *separable state*.

Unlike this type of states that could be represented as a tensor product of two states in $\mathcal{H}_1$ and $\mathcal{H}_2$, we have *Bell states*, also known as *EPR pairs*. Bell states are four specific maximally entangled quantum states of two qubits that represent one of the simplest examples of quantum entanglement.

$$\begin{aligned} |\Phi^+\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle) \\ |\Phi^-\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\uparrow\rangle - |\downarrow\rangle \otimes |\downarrow\rangle) \\ |\Psi^+\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\downarrow\rangle + |\downarrow\rangle \otimes |\uparrow\rangle) \\ |\Psi^-\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \end{aligned}$$

Example: (Maximally Entangled State: $|\Phi^+\rangle$)

Consider a maximally entangled state of two qubits,

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle)$$

To prove that this is an entangled pure state, we must show that it cannot be written as a simple tensor product. We proceed by contradiction. Assume that $|\Phi^+\rangle$ is separable. Then, there must exist general single-qubit states $|\psi_1\rangle \in \mathcal{H}_1$ and $|\psi_2\rangle \in \mathcal{H}_2$ such that $|\Phi^+\rangle = |\psi_1\rangle \otimes |\psi_2\rangle$.

Let us express $|\psi_1\rangle$ and $|\psi_2\rangle$ in the standard spin- z basis:

$$\begin{aligned} |\psi_1\rangle &= a |\uparrow\rangle + b |\downarrow\rangle \\ |\psi_2\rangle &= c |\uparrow\rangle + d |\downarrow\rangle \end{aligned}$$

where a, b, c, d are complex coefficients. Taking their tensor product yields:

$$\begin{aligned} |\psi_1\rangle \otimes |\psi_2\rangle &= (a |\uparrow\rangle + b |\downarrow\rangle) \otimes (c |\uparrow\rangle + d |\downarrow\rangle) \\ &= ac(|\uparrow\rangle \otimes |\uparrow\rangle) + ad(|\uparrow\rangle \otimes |\downarrow\rangle) + bc(|\downarrow\rangle \otimes |\uparrow\rangle) + bd(|\downarrow\rangle \otimes |\downarrow\rangle) \end{aligned}$$

For this separable state to equal our target state $|\Phi^+\rangle$, the coefficients of the basis vectors must

match exactly. Comparing the terms gives us the following system of equations:

$$ac = \frac{1}{\sqrt{2}} \quad (1)$$

$$ad = 0 \quad (2)$$

$$bc = 0 \quad (3)$$

$$bd = \frac{1}{\sqrt{2}} \quad (4)$$

From equation (2), $ad = 0$, which implies that either $a = 0$ or $d = 0$.

- If $a = 0$, then equation (1) becomes $0 = \frac{1}{\sqrt{2}}$, which is false.
- If $d = 0$, then equation (4) becomes $0 = \frac{1}{\sqrt{2}}$, which is also false.

Since both possibilities lead to a mathematical contradiction, our initial assumption must be incorrect. Therefore, the state cannot be factored into the form $|\psi_1\rangle \otimes |\psi_2\rangle$. Hence, $|\Phi^+\rangle$ is an entangled pure state.

Example: (The EPR-Bohm Pair: $|\Psi^-\rangle$)

Consider another maximally entangled state, also known as the singlet state:

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$$

Assume, that this state is separable and can be written as $|\psi_1\rangle \otimes |\psi_2\rangle$. Using the same general single-qubit states as before:

$$\begin{aligned} |\psi_1\rangle \otimes |\psi_2\rangle &= (a|\uparrow\rangle + b|\downarrow\rangle) \otimes (c|\uparrow\rangle + d|\downarrow\rangle) \\ &= ac(|\uparrow\rangle \otimes |\uparrow\rangle) + ad(|\uparrow\rangle \otimes |\downarrow\rangle) + bc(|\downarrow\rangle \otimes |\uparrow\rangle) + bd(|\downarrow\rangle \otimes |\downarrow\rangle) \end{aligned}$$

Equating the coefficients of this expansion to the coefficients of our target singlet state gives the following system of equations:

$$ac = 0 \quad (5)$$

$$ad = \frac{1}{\sqrt{2}} \quad (6)$$

$$bc = -\frac{1}{\sqrt{2}} \quad (7)$$

$$bd = 0 \quad (8)$$

From equation (5), $ac = 0$, which implies that either $a = 0$ or $c = 0$.

- If $a = 0$, then equation (6) becomes $0 = \frac{1}{\sqrt{2}}$, which is false.
- If $c = 0$, then equation (7) becomes $0 = -\frac{1}{\sqrt{2}}$, which is also false.

Once again, assuming the state to be separable leads to a mathematical contradiction. Hence, the singlet state $|\Psi^-\rangle$ is also an entangled pure state.

C. The Einstein, Podolsky, Rosen Paradox

In the year 1935, *Einstein, Podolsky* and *Rosen* (EPR) attempted to use the property of entanglement to argue that quantum mechanics is an “incomplete theory”. Their argument involves preparing two particles in an EPR state ($|\Psi^-\rangle$) and subsequently separating those particles to a large distance between two observers, *Alice* and *Bob*.

The EPR thought experiment:

- We prepare two particles in an EPR state, $|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle)$
- Separate the pair to a large distance across two distinct observers, *Alice* and *Bob*.
- If *Alice* chooses to measure the spin of her particle in the z -direction. There are two distinct possibilities, either she finds spin-up $|\uparrow\rangle$ or spin-down $|\downarrow\rangle$ each with a 50% probability.
- As soon as *Alice* measures her particle, the wavefunction of the combined state collapses instantaneously. If she finds spin-up ($|\uparrow\rangle$), that means $|\Psi^-\rangle$ collapsed to ($|\uparrow\rangle \otimes |\downarrow\rangle$). In that case, *Bob* must find spin-down ($|\downarrow\rangle$). Similarly, if *Alice* found spin-down ($|\downarrow\rangle$) then *Bob* must find spin-up ($|\uparrow\rangle$).
- No matter who measured first, the result must be *anti-correlated*. This correlation appears due to the instantaneous collapse of the wavefunction.
- **The Principle of Locality:** According to the special theory of relativity, no information or physical influence can travel faster than the speed of light c . Because *Alice* and *Bob* are separated by a large distance, *Alice*’s measurement cannot physically send a signal to affect *Bob*’s particle instantaneously.
- So, *Bob*’s particle must be in a definite state as soon as *Alice* makes a measurement, and no physical signal from *Alice* could have reached *Bob*’s particle.

This thought experiment sparks immense tension with classical intuition. In the relativistic framework, our universe is strictly local. No physical influence can travel faster than the speed of light. But quantum entanglement shatters this framework. The fact that measuring one entangled particle instantaneously determines the state of its partner, no matter the distance between them, forces us to confront what Einstein famously called “spooky action at a distance”. It creates a philosophical and physical conflict. We must either abandon our deeply ingrained classical assumption that the universe is strictly local or accept that quantum mechanics represents a fundamentally incomplete picture of a reality we do not yet fully grasp.

D. Hidden variables

In the case of quantum entanglement, we found an anti-correlation relationship between the measurements of entangled particles. By measuring her particle, *Alice* instantaneously knows the spin of *Bob's* particle in the z -direction.

What if *Alice* and *Bob* choose to measure along a different axis? Choose any arbitrary direction on the x - z plane $a = (\sin \theta, 0, \cos \theta)$. Let's find out the operator σ_θ to measure spin along a .

$$\sigma_\theta = \sigma \cdot a = \sigma_x \sin \theta + \sigma_z \cos \theta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \sin \theta + \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \cos \theta$$

$$\therefore \sigma_\theta = \begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$$

Eigenvectors of σ_θ are,

$$\begin{aligned} |\theta_+\rangle &= \cos \frac{\theta}{2} |\uparrow\rangle + \sin \frac{\theta}{2} |\downarrow\rangle \\ |\theta_-\rangle &= \cos \frac{\theta}{2} |\downarrow\rangle - \sin \frac{\theta}{2} |\uparrow\rangle \end{aligned}$$

As $\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} = 1$, we have,

$$\begin{aligned} |\uparrow\rangle &= \cos \frac{\theta}{2} |\theta_+\rangle - \sin \frac{\theta}{2} |\theta_-\rangle \\ |\downarrow\rangle &= \sin \frac{\theta}{2} |\theta_+\rangle + \cos \frac{\theta}{2} |\theta_-\rangle \\ \therefore |\Psi^-\rangle &= \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) \\ &= \frac{1}{\sqrt{2}} (|\theta_+\rangle \otimes |\theta_-\rangle - |\theta_-\rangle \otimes |\theta_+\rangle) \end{aligned}$$

So, we find that for measurement along any arbitrary but same axis, *Alice* and *Bob* always find anti-correlated spins. To explain this “perfect correlation” between both observers’ measurements, Einstein, Podolsky and Rosen argued that particles must carry some pre-existing objective properties that dictate the outcome. These pre-determined properties are referred as *local hidden variables*. Hidden variables function like pre defined instructions that determine the results of the measurements of each particle.

Hidden variables (λ): Hypothetical, unobservable parameters postulated to supplement the quantum wavefunction $|\psi\rangle$ to provide a complete, deterministic description of a physical system. Formally, they assert that the definite outcome of any measurement is governed by a deterministic function $A(a, \lambda)$ depending solely on the local measurement setting a and the hidden state $\lambda \in \Lambda$. In this framework, the probabilistic predictions of standard quantum mechanics are simply classical statistical averages over an unknown probability density function $\rho(\lambda)$ of these parameters.

Incompleteness of Quantum theory: Since the standard formulation of quantum mechanics relies on the wave function, which only provides probabilities for these outcomes and does not account for any pre-existing hidden variables. Therefore, EPR concluded that the quantum mechanical description of physical reality, which is represented by the wavefunction, must be incomplete.

The Problem of Counterfactual reasoning: Counterfactual reasoning is a process of asking ‘what if’ questions that have not actually occurred. Einstein, Podolsky, and Rosen used counterfactual reasoning to conclude that these pre-existing instructions (local hidden variables) must exist. They believed that because we can imagine asking the particle different questions, the particle must have the answers to all those questions written down inside it, even the questions we didn’t ask. By introducing *hidden variables*, EPR advocated for *local realism*. That is, any observation made is a result of some predefined properties, and these properties must be local.

However, almost three decades later, *Bell’s theorem* proved that applying counterfactual reasoning in quantum mechanics leads to mathematical inequalities that quantum mechanics violates. We simply have to admire that **unperformed experiments have no results**.

E. Bell's inequality

To illustrate the limitations of hidden variable theories, consider a scenario where *Alice* and *Bob* are given a choice to measure spin along a third possible axis. Each can measure the spin along the z -axis, along the $a = (\sin \theta, 0, \cos \theta)$ axis, or along the $b = (\sin \phi, 0, \cos \phi)$ axis. If local hidden variables exist, each particle must be assigned a hidden variable that determines the choice of each of these measurements.

Suppose Alice measures her spin along the z -axis and Bob measures his spin along an arbitrary axis $a = (\sin \theta, 0, \cos \theta)$. Start with the singlet state

$$|\Psi^-\rangle = \frac{1}{\sqrt{2}} (|\uparrow\rangle \otimes |\downarrow\rangle - |\downarrow\rangle \otimes |\uparrow\rangle) = \frac{1}{\sqrt{2}} (|\theta_+\rangle \otimes |\theta_-\rangle - |\theta_-\rangle \otimes |\theta_+\rangle)$$

We can now calculate the conditional probabilities for Bob's measurements:

- **If Alice measures $\sigma_z^A = +$ (Bob's state is $|\downarrow\rangle_B$):**

The probability Bob measures $+$ is $|\langle\theta_+|\downarrow\rangle|^2 = \sin^2 \frac{\theta}{2}$.

The probability Bob measures $-$ is $|\langle\theta_-|\downarrow\rangle|^2 = \cos^2 \frac{\theta}{2}$.

- **If Alice measures $\sigma_z^A = -$ (Bob's state is $|\uparrow\rangle_B$):**

The probability Bob measures $+$ is $|\langle\theta_+|\uparrow\rangle|^2 = \cos^2 \frac{\theta}{2}$.

The probability Bob measures $-$ is $|\langle\theta_-|\uparrow\rangle|^2 = \sin^2 \frac{\theta}{2}$.

To find the joint probabilities $P(\sigma_z^A, \sigma_\theta^B)$, we multiply these conditional probabilities by the 50% probability of Alice's initial measurement:

$$\begin{aligned} P(\sigma_z^A = +, \sigma_\theta^B = +) &= \frac{1}{2} \sin^2 \frac{\theta}{2} \\ P(\sigma_z^A = +, \sigma_\theta^B = -) &= \frac{1}{2} \cos^2 \frac{\theta}{2} \\ P(\sigma_z^A = -, \sigma_\theta^B = +) &= \frac{1}{2} \cos^2 \frac{\theta}{2} \\ P(\sigma_z^A = -, \sigma_\theta^B = -) &= \frac{1}{2} \sin^2 \frac{\theta}{2} \end{aligned}$$

To formulate a local hidden variable theory, we assume each particle carries pre-determined labels (hidden variables) that dictate the outcome of any measurement.

For the EPR state, if Alice and Bob measure along the exact same axis ($\theta = 0$), their results are perfectly anti-correlated. To reproduce this perfect anti-correlation using hidden variables, the value of Bob's hidden variables must be the exact opposite of Alice's. Therefore, we must assign:

$$s_z^B = -s_z^A \quad \text{and} \quad s_\theta^B = -s_\theta^A$$

Because of this rigid anti-correlation, if Bob measures his spin to be -1 along the θ axis ($\sigma_\theta^B = -1$), it strictly implies that Alice's particle must possess the hidden variable $s_\theta^A = +1$.

By substituting $\sigma_\theta^B = -s_\theta^A$ into our quantum mechanical joint probabilities, we can deduce the marginal probability distribution $p(s_z^A, s_\theta^A)$ for the hidden variables of Alice's particle alone:

- $P(s_z^A = +, s_\theta^A = +) = P(\sigma_z^A = +, \sigma_\theta^B = -) = \frac{1}{2} \cos^2 \frac{\theta}{2}$
- $P(s_z^A = +, s_\theta^A = -) = P(\sigma_z^A = +, \sigma_\theta^B = +) = \frac{1}{2} \sin^2 \frac{\theta}{2}$
- $P(s_z^A = -, s_\theta^A = +) = P(\sigma_z^A = -, \sigma_\theta^B = -) = \frac{1}{2} \sin^2 \frac{\theta}{2}$
- $P(s_z^A = -, s_\theta^A = -) = P(\sigma_z^A = -, \sigma_\theta^B = +) = \frac{1}{2} \cos^2 \frac{\theta}{2}$

These are the exact marginal distributions required for a hidden variable theory to account for the measurement probabilities of an EPR pair.

This means Alice's particle comes with variables s_z^A, s_θ^A and s_ϕ^A , each of which can take a value of ± 1 . The probabilities of the different choices would then be governed by some joint probability distribution $p(s_z^A, s_\theta^A, s_\phi^A)$

Let's assume that such a distribution does exist, which implies certain relations between the probability distributions $P(s_i^A, s_j^A)$. For example, by summing over the variables which were not measured, we find:

$$P(s_\theta^A = +, s_\phi^A = -) = p(++-) + p(-+-)$$

Because probabilities are always non-negative, we can establish an inequality by adding further positive probability terms:

$$\begin{aligned} p(++-) + p(-+-) &\leq [p(++-) + p(+++)] + [p(-+-) + p(---)] \\ \implies P(s_\theta^A = +, s_\phi^A = -) &= P(s_z^A = +, s_\theta^A = +) + P(s_z^A = -, s_\theta^A = -) \end{aligned}$$

We know what each of these marginal distributions must be from the quantum mechanical predictions for the EPR state. Substituting those known distributions into this relation gives the Bell inequality:

$$\sin^2 \frac{\theta - \phi}{2} \leq \cos^2 \frac{\theta}{2} + \cos^2 \frac{\phi}{2}$$

where the left-hand side follows from the rotational invariance of the EPR state.

Violation of Bell's Inequality:

There is a fundamental problem with the Bell inequality: it is simply not true for all values of θ and ϕ . Suppose, for example, that we take $\theta = 3\pi/2$ and $\phi = 3\pi/4$. Evaluating the components yields:

$$\sin^2 \frac{3\pi}{8} - \cos^2 \frac{3\pi}{8} = -\cos \frac{3\pi}{4} = \frac{1}{\sqrt{2}}$$

Meanwhile, evaluating the term for the right-hand side gives:

$$\cos^2 \frac{3\pi}{4} = \frac{1}{2}$$

Obviously, $1/2 < 1/\sqrt{2}$, which shows that these specific measurement angles violate the Bell inequality.

F. CHSH inequality

Once again, assume *Alice* and *Bob* are sent a two state quantum system. *Alice* can choose to measure either A_1 or A_2 and similarly, *Bob* can choose to measure B_1 or B_2 . We can think of these observables as the spins along different axes. Every observable has 2 possible eigenvalues, $a_i = \pm 1$ and $b_i = \pm 1$ ($i = 1, 2$).

For hidden variable theory: Let's find out the expectation value of the observable,

$$C = (A_1 + A_2)B_1 + (A_1 - A_2)B_2$$

If the system has well defined hidden variables and follows local realism, then measurements of A_i and B_i should be independent of each other. So,

$$C_{hv} = (a_1 + a_2)b_1 + (a_1 - a_2)b_2$$

Now, $a_1, a_2 = \pm 1 \implies (a_1 + a_2) = \pm 2$ and $(a_1 - a_2) = \pm 2$

In either case, $C_{hv} = \pm b_i$ ($i = 1, 2$).

Since b_i can only take values in ± 1 , we have $|\langle b_i | b_i \rangle| \leq 1$

$$\therefore -2 \leq \langle C_{hv} | C_{hv} \rangle \leq +2$$

This is CHSH inequality on classical correlation.

For quantum mechanics: In quantum theory, we must manipulate C as an operator equation. Squaring C yields.

$$\begin{aligned} C^2 &= 4\mathbb{I} + A_1 A_2 B_1 B_2 + A_2 A_1 B_1 B_2 + A_1 A_2 B_2 B_1 - A_2 A_1 B_2 B_1 \\ &= 4\mathbb{I} - [A_1, A_2][B_1, B_2] \end{aligned}$$

Now,

$$|\langle [A_1, A_2][A_1, A_2] \rangle| \leq |\langle A_1 A_2 | A_1 A_2 \rangle| + |\langle A_2 A_1 | A_2 A_1 \rangle| \leq 1+1 = 2 \quad \implies |\langle [A_1, A_2][A_1, A_2] \rangle| \leq 2$$

Applying the same logic, $|\langle [B_1, B_2][B_1, B_2] \rangle| \leq 2$

$$\therefore 0 \leq \langle C^2 | C^2 \rangle \leq 8$$

As, $\langle C^2 | C^2 \rangle \geq \langle C | C \rangle^2$,

$$\begin{aligned} 0 &\leq \langle C | C \rangle^2 \leq 8 \\ \implies -2\sqrt{2} &\leq \langle C | C \rangle \leq 2\sqrt{2} \end{aligned}$$

Classical hidden variable theories are restricted to $-2 \leq \langle C | C \rangle \leq 2$. Quantum mechanics allows for stronger correlations. From experimental proofs of a stronger correlation, we can prove the nonexistence of hidden variables.

G. The CHSH Game

The CHSH inequality can be intuitively operationalized as a cooperative thought experiment played by two spatially separated observers against a referee. This framework, known as the *CHSH game*, provides a clear, quantifiable demonstration of the gap between classical local realism and quantum mechanics.

Rules of the Game:

- A referee generates two classical bits chosen uniformly at random: $x \in 0, 1$ and $y \in 0, 1$.

- The referee sends x to *Alice* and y to *Bob*.
- Once the bits are distributed, no communication is permitted between *Alice* and *Bob* (enforcing the principle of locality).
- Upon receiving their respective bits, *Alice* must output a bit $a \in 0, 1$ and *Bob* must output a bit $b \in 0, 1$.
- **Winning Condition:** They win the game if and only if their outputs satisfy the boolean equation: $a \oplus b = x \cdot y$

where $\oplus$ represents addition modulo 2 (XOR), and $\cdot$ represents logical AND.

According to this winning condition, if $x = 1$ and $y = 1$, *Alice* and *Bob* must output different bits ($a \neq b$). For all other combinations of inputs $((0, 0), (0, 1), (1, 0))$, they must output the same bit ($a = b$).

The Classical Limit: Assume *Alice* and *Bob* rely entirely on classical physics. Before the game starts, they can agree on any deterministic or probabilistic strategy, which may include shared *local hidden variables* (λ).

A simple deterministic strategy would be for both to agree to always output 0 ($a = 0, b = 0$). In this scenario, $a \oplus b = 0$ is always true, meaning they win whenever $x \cdot y = 0$. Because the inputs are perfectly random, this happens in 3 out of the 4 possible scenarios, yielding a 75% win rate.

Can they do better by utilizing complex hidden variables? Because no pre-existing set of local instructions can simultaneously satisfy all four conditions of the game without causing a logical contradiction, Bell's theorem guarantees that no classical strategy can exceed this 75% threshold. The maximum classical win probability is rigidly bounded:

$$P_{\text{classical}} \leq 0.75$$

The Quantum Strategy: Quantum entanglement provides a path to surpass this classical limit. *Alice* and *Bob* can share a maximally entangled state, such as $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|\uparrow\rangle \otimes |\uparrow\rangle + |\downarrow\rangle \otimes |\downarrow\rangle)$, before they are separated.

They agree on a strategy utilizing the spin operator σ_θ derived in the previous section. They map their classical outputs $a, b \in 0, 1$ directly to their measurement results: outputting 0 if they measure a +1 eigenvalue (the $|\theta_+\rangle$ state) and 1 if they measure a -1 eigenvalue (the $|\theta_-\rangle$ state).

Alice chooses her measurement axis based on her bit x :

- If $x = 0$, she measures along the z -axis ($\theta_A = 0^\circ$).
- If $x = 1$, she measures along the x -axis ($\theta_A = 90^\circ$).

Bob chooses his measurement axis based on his bit y :

- If $y = 0$, he measures at an angle $\theta_B = 45^\circ$.
- If $y = 1$, he measures at an angle $\theta_B = -45^\circ$.

By carefully misaligning their measurement axes by 45° , the expectation value of their joint measurements leverages the expanded quantum correlation bound of $-2 \leq \langle C|C \rangle \leq 2$ rather than the strict classical $[-2, 2]$ constraint. Calculating the precise probabilities for all four input combinations reveals that this quantum strategy achieves an optimal win rate known as *Tsirelson's bound*:

$$P_{\text{quantum}} = \cos^2\left(\frac{\pi}{8}\right) = \frac{1}{2} + \frac{1}{2\sqrt{2}} \approx 0.853$$

By winning approximately 85.3% of the time, *Alice* and *Bob* probably outperform any local hidden variable theory. The CHSH game therefore frames quantum non-locality not merely as a philosophical paradox, but as an observable, quantifiable operational advantage.

H. The No-Cloning Theorem

In classical computing, one of the fundamental operations is *copying data*. One can copy as classical bit as many times as needed. However, in the quantum world, it is physically impossible to create an identical and independent copy of a quantum state. We simply lack a physical evolution (an unitary operator) that can clone an arbitrary quantum state.

No-Cloning Theorem: There exists no unitary operator U that can take an arbitrary unknown quantum state $|\psi\rangle$ and a blank initialized target state $|e\rangle$, and output two identical copies of $|\psi\rangle$. *The transformation $U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle$ is not possible for any arbitrary state $|\psi\rangle$ and $|e\rangle$.*

Proof by Contradiction:

Suppose there exists a unitary operator U . Let this operator act on a combined system consisting of the state we want to copy and a standard “blank” state $|e\rangle$.

If U is a universal “cloning operator”, it must successfully copy any two arbitrary states, say $|\psi\rangle$ and $|\phi\rangle$. Therefore, the operator must satisfy:

$$U(|\psi\rangle \otimes |e\rangle) = |\psi\rangle \otimes |\psi\rangle$$

$$U(|\phi\rangle \otimes |e\rangle) = |\phi\rangle \otimes |\phi\rangle$$

Now, let's take the inner product of these two equations. Evaluate the adjoint of the first equation and multiply it by the second:

$$(\langle\psi| \otimes \langle e|)U^\dagger U(|\phi\rangle \otimes |e\rangle) = (\langle\psi| \otimes \langle\psi|)(|\phi\rangle \otimes |\phi\rangle)$$

Because U is a valid physical evolution, it must be unitary. $\implies U^\dagger U = \mathbb{I}$

$$\therefore (\langle\psi| \otimes \langle e|)\mathbb{I}(|\phi\rangle \otimes |e\rangle) = \langle\psi|\phi\rangle\langle\psi|\phi\rangle \langle e|e\rangle\langle e|e\rangle$$

Since $|e\rangle$ is a valid, normalized quantum state, $\langle e|e\rangle = 1$.

For the right side of the equation, use the tensor product property for inner products, $(\langle A| \otimes \langle B|)(|C\rangle \otimes |D\rangle) = \langle A|C\rangle\langle B|D\rangle$:

$$(\langle\psi| \otimes \langle\psi|)(|\phi\rangle \otimes |\phi\rangle) = \langle\psi|\phi\rangle\langle\psi|\phi\rangle \langle\psi|\phi\rangle\langle\psi|\phi\rangle = \langle\psi|\phi\rangle\langle\psi|\phi\rangle^2$$

Equating LHS and RHS:

$$\langle\psi|\phi\rangle\langle\psi|\phi\rangle = \langle\psi|\phi\rangle\langle\psi|\phi\rangle^2$$

The algebraic equation, $x = x^2$, has only two possible mathematical solutions: $x = 0$ or $x = 1$.

- If $\langle\psi|\phi\rangle\langle\psi|\phi\rangle = 1$, the states $|\psi\rangle$ and $|\phi\rangle$ are identical.
- If $\langle\psi|\phi\rangle\langle\psi|\phi\rangle = 0$, the states $|\psi\rangle$ and $|\phi\rangle$ are perfectly orthogonal.

So, we have a mathematical contradiction to our initial assumption that U can clone *arbitrary* unknown states. The U operator can only successfully copy states if they are chosen from a pre-defined set of mutually orthogonal basis vectors. Therefore, a universal quantum cloning machine cannot exist.

Implications for Entanglement and Locality:

The no cloning theorem is very important safeguard to prevent “spooky action at a distance”. Assume, if *Alice* and *Bob* share an EPR pair $|\Psi^-\rangle$. Alice measures her particle, and so the combined

wavefunction collapses instantaneously. If Bob were able to clone his single, unknown particle, he could create millions of copies, measure them to determine the exact state they collapsed into. And use that information to communicate instantaneously. By strictly forbidding the copying of unknown states, the no-cloning theorem ensures that quantum entanglement cannot be used for faster-than-light communication.

II. SUPERDENSE CODING

A. Premise

The idea of superdense coding was first proposed in the seminal work of Bennett and Wiesner [1] in 1984. The premise of superdense coding is to use entanglement (preshared) as a resource so that fewer qubits of communication are needed to transmit a given number of classical bits. In this context, Holevo's bound [2] provides the relevant benchmark: If sender (Alice) encodes a classical message into an ensemble of states on an n -qubit system, sends that system to a receiver (Bob), and receiver (Bob) has no extra side information correlated with the message. Then Bob can learn no more than n bits of information by means of the process he and Alice have performed [3].

The immediate question that arises is whether this one bit per one qubit limit can be overcome by supplying additional quantum resources. Superdense coding shows that it can in some sense: with one preshared entangled pair (recall we showed in section I G that entanglement can be used as a resource), a single transmitted qubit suffices to convey two classical bits. However, it is worth to stress on the fact that the advantage of using fewer communication cost (qubits) is already paid up for by subscribing to an additional resource i.e one preshared ebit, which must be distributed beforehand. So superdense coding is in some sense 'dense' if we can afford to have the established resource.

$$1 \text{ transmitted qubit} + 1 \text{ shared ebit} \implies 2 \text{ classical bits.}$$

B. Quantum gates used in the protocol

Before going into the protocol, we mention the unitary gates that are used in the protocol.

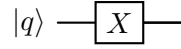
Identity gate I

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I|0\rangle = |0\rangle, \quad I|1\rangle = |1\rangle.$$

$$|q\rangle \text{ --- } \boxed{I} \text{ ---}$$

Pauli-X gate

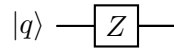
$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad X|0\rangle = |1\rangle, \quad X|1\rangle = |0\rangle.$$



This is the bit flip operator.

Pauli-Z gate

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad Z|0\rangle = |0\rangle, \quad Z|1\rangle = -|1\rangle.$$

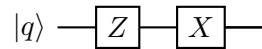


so it flips the sign of the $|1\rangle$ amplitude.

Combined operation XZ

Here XZ means apply Z first and then X

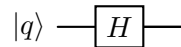
$$XZ = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad XZ|0\rangle = |1\rangle, \quad XZ|1\rangle = -|0\rangle.$$



Up to a global phase, XZ is equivalent to the Pauli-Y operation ($XZ = -iY$). This is both a phase flip and a bit flip.

Hadamard gate H

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H|0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad H|1\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}.$$



This maps computational basis states to X-basis states, i.e $|+\rangle, |-\rangle$

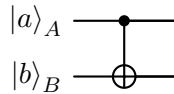
Controlled-NOT gate (CNOT) All of the above discussed are single unitary gates. The CNOT is a two-qubit unitary gate. The CNOT acts on two qubits. If the control qubit is $|1\rangle$, the target qubit is flipped.

$$|b\rangle_B |a\rangle_A \longrightarrow |b \oplus a\rangle_B |a\rangle_A.$$

In this section, the joint state is ordered as

$$|ba\rangle_{BA} = |b\rangle_B \otimes |a\rangle_A,$$

so the first slot is Bob's qubit and the second slot is Alice's qubit.



Although the circuit is drawn with Alice on top and Bob below, throughout this section we write the joint state in the order BA.

C. The protocol

The protocol consists of four steps: preparation and sharing of an entangled pair, encoding, transmission, and decoding. Figure 1 provides an overview of the complete procedure, while Fig. 2 presents the detailed circuit level implementation.

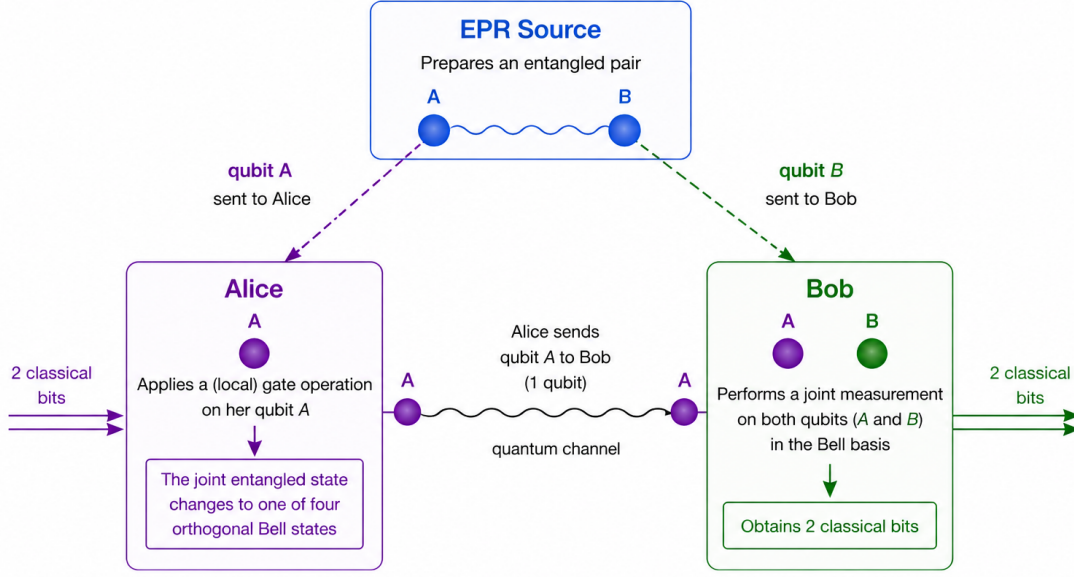


FIG. 1. Schematic illustration of the superdense coding protocol. An entangled pair is distributed between Alice and Bob. Alice encodes two classical bits by applying a local operation on her qubit and sends it to Bob, who performs a joint measurement to recover the message.

1. 1. Entanglement (state preparation)

Before any message is sent, a source (the third party agency to whom Alice and Bob subscribed for ebit!) prepares an entangled EPR pair and distributes one qubit to Alice and the other to Bob.

Mathematically the preparation is as follows : We start with two $|0\rangle$ qubits.

$$|0\rangle_B \otimes |0\rangle_A.$$

First apply a Hadamard gate to Alice's qubit:

$$(I_B \otimes H_A)(|0\rangle_B \otimes |0\rangle_A) = |0\rangle_B \otimes \frac{|0\rangle_A + |1\rangle_A}{\sqrt{2}} = \frac{|00\rangle_{BA} + |01\rangle_{BA}}{\sqrt{2}}.$$

Then apply a CNOT with control on A and target on B :

$$|b\rangle_B |a\rangle_A \rightarrow |b \oplus a\rangle_B |a\rangle_A.$$

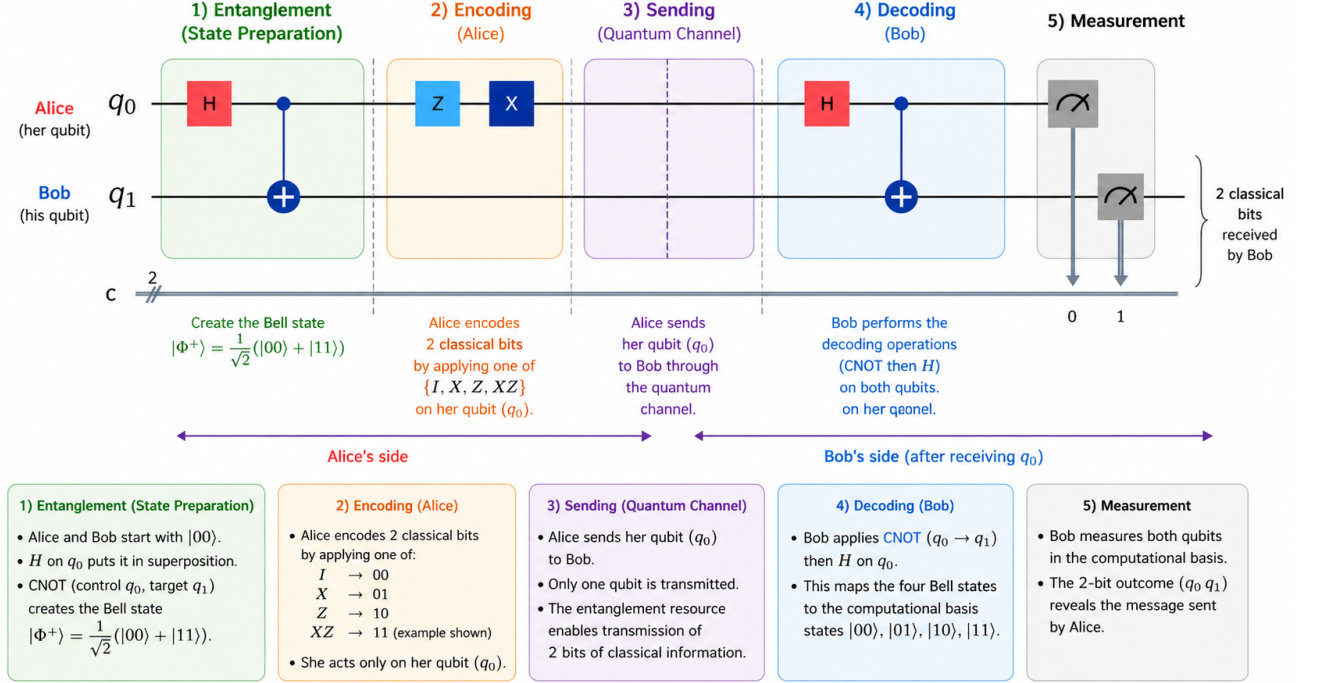


FIG. 2. Quantum circuit level implementation of the superdense coding protocol.

So the two basis terms $|00\rangle_{BA}$ and $|01\rangle_{BA}$ evolve as

$$|00\rangle_{BA} \rightarrow |00\rangle_{BA}, \quad |01\rangle_{BA} \rightarrow |11\rangle_{BA}.$$

Therefore the shared state becomes

$$|\Phi^+\rangle_{BA} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Now he shares each respective qubits to both. At this point Alice holds qubit A and Bob holds qubit B .

2. 2. Encoding (Alice)

Alice wants to send two classical bits. She does this by applying one of four local unitary operations to her qubit:

$$\{I, X, Z, XZ\}.$$

Note, the operation acts only on Alice's side. Because the pair is entangled, that local action changes the *global* state of the two qubit system.

Recall, The four Bell states (written in the BA ordering) are

$$\begin{aligned} |\Phi^+\rangle_{BA} &= \frac{|00\rangle + |11\rangle}{\sqrt{2}}, & |\Phi^-\rangle_{BA} &= \frac{|00\rangle - |11\rangle}{\sqrt{2}}, \\ |\Psi^+\rangle_{BA} &= \frac{|01\rangle + |10\rangle}{\sqrt{2}}, & |\Psi^-\rangle_{BA} &= \frac{|01\rangle - |10\rangle}{\sqrt{2}}. \end{aligned}$$

These four states are mutually orthogonal and form an orthonormal basis for the two qubit Hilbert space.

Now to encode four different messages, Alice has to apply the four possible local operations to the initial state $|\Phi^+\rangle_{BA}$:

(i) Identity

$$(I \otimes I) |\Phi^+\rangle_{BA} = |\Phi^+\rangle_{BA}.$$

(ii) Phase flip on Alice

$$(I \otimes Z) |\Phi^+\rangle_{BA} = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\Phi^-\rangle_{BA}.$$

(iii) Bit flip on Alice

$$(I \otimes X) |\Phi^+\rangle_{BA} = \frac{|01\rangle + |10\rangle}{\sqrt{2}} = |\Psi^+\rangle_{BA}.$$

(iv) Bit and phase flip on Alice

$$(I \otimes XZ) |\Phi^+\rangle_{BA} = \frac{|01\rangle - |10\rangle}{\sqrt{2}} = |\Psi^-\rangle_{BA}.$$

So Alice's two-bit message is mapped to one of four *orthogonal* Bell states:

$$00 \mapsto |\Phi^+\rangle, \quad 01 \mapsto |\Phi^-\rangle, \quad 10 \mapsto |\Psi^+\rangle, \quad 11 \mapsto |\Psi^-\rangle.$$

Note: She is not encoding two bits into a single isolated qubit; rather, she selects one of four orthogonal states of the joint two qubit system, each corresponding to a distinct classical message. And this choice of correspondence is arbitrary but must be agreed beforehand by Bob and Alice.

Why orthogonality matters ? [4]

Suppose a set of states $\{|\psi_i\rangle\}$ is orthonormal. Then one can define measurement operators

$$M_i = |\psi_i\rangle \langle \psi_i|,$$

which satisfy the completeness relation

$$\sum_i M_i = I.$$

. If the system is prepared in the state $|\psi_i\rangle$, the probability of obtaining outcome i is

$$p(i) = \langle \psi_i | M_i | \psi_i \rangle = 1,$$

so the states *can be distinguished with certainty*. By contrast, if the states $|\psi_i\rangle$ are not orthonormal then it can be proved [4] that there is *no quantum measurement capable of distinguishing the states*. [4].

In superdense coding, the four Bell states form such an orthonormal set:

$$\{|\Phi^\pm\rangle, |\Psi^\pm\rangle\}.$$

A Bell basis measurement is precisely the projective measurement defined by the operators

$$\{|\Phi^\pm\rangle\langle\Phi^\pm|, |\Psi^\pm\rangle\langle\Psi^\pm|\}.$$

Because these states are mutually orthogonal, this measurement allows Bob to determine exactly which state was sent to him and hence which classical message was prepared.

3. 3. Sending

Alice now sends her qubit A to Bob. This is the only physical transmission in the protocol. Once Bob receives it, he has both qubits and can perform *joint measurement* i.e., a measurement acting on the composite Hilbert space $\mathcal{H}_B \otimes \mathcal{H}_A$.

A small aside: Before this step, Bob possesses only his subsystem B , whose reduced density operator is (calculation is shown in the next section)

$$\rho_B = \text{Tr}_A(\rho_{BA}) = \frac{I}{2},$$

which is independent of Alice's encoded qubit. Hence, prior to receiving qubit A , any measurement performed by Bob on his subsystem alone yields no information about the encoded message.

After transmission, Bob holds both qubits B and A , and the system is in one of the four Bell states

$$\{|\Phi^\pm\rangle_{BA}, |\Psi^\pm\rangle_{BA}\},$$

depending on Alice's encoding.

4. 4. Decoding (Bob)

Once Bob has both qubits, he can perform a joint measurement on the composite system BA . Since the global state is one of four mutually orthogonal Bell states, ideally, Bob could directly measure in the Bell basis (as discussed above [II C 2](#)). However, most physical platforms perform measurements in the computational basis. To implement the Bell measurement, Bob applies a unitary transformation U (a CNOT gate followed by a Hadamard on Alice's qubit) that maps the Bell states to the computational basis, and then performs a standard computational basis measurement.

The decoding circuit is:

- a CNOT with control on A and target on B ,
- followed by a Hadamard gate on qubit A .

Note, The certainty of the measurement that was established in the previous section remains intact and it simply follows from the unitarity of U (a CNOT gate followed by a Hadamard on Alice's qubit). Since unitary transformations preserve inner products, the orthogonality of the Bell states is preserved under U , and each state is mapped to a distinct computational basis vector. Therefore, the measurement outcomes occur with certainty:

$$p(i) = \langle \psi_i | U^\dagger (|i\rangle \langle i|) U | \psi_i \rangle = 1,$$

where $\{|\psi_i\rangle\}$ denotes the Bell states and $\{|i\rangle\}$ the computational basis.

Now Bob applies unitary transformation U on the four Bell basis states and map them to computational basis.

(i) $|\Phi^+\rangle_{BA}$

$$|\Phi^+\rangle_{BA} = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

After CNOT ($A \rightarrow B$):

$$\frac{|00\rangle + |01\rangle}{\sqrt{2}}.$$

After Hadamard on A :

$$|00\rangle.$$

(ii) $|\Phi^-\rangle_{BA}$

$$|\Phi^-\rangle_{BA} = \frac{|00\rangle - |11\rangle}{\sqrt{2}}.$$

After CNOT:

$$\frac{|00\rangle - |01\rangle}{\sqrt{2}}.$$

After Hadamard on A :

$$|01\rangle.$$

(iii) $|\Psi^+\rangle_{BA}$

$$|\Psi^+\rangle_{BA} = \frac{|01\rangle + |10\rangle}{\sqrt{2}}.$$

After CNOT:

$$\frac{|11\rangle + |10\rangle}{\sqrt{2}}.$$

After Hadamard on A :

$$|10\rangle.$$

(iv) $|\Psi^-\rangle_{BA}$

$$|\Psi^-\rangle_{BA} = \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

After CNOT:

$$\frac{|11\rangle - |10\rangle}{\sqrt{2}}.$$

After Hadamard on A :

$$|11\rangle.$$

Hence the decoding circuit maps the Bell basis to the computational basis:

$$|\Phi^+\rangle \rightarrow |00\rangle, \quad |\Phi^-\rangle \rightarrow |01\rangle,$$

$$|\Psi^+\rangle \rightarrow |10\rangle, \quad |\Psi^-\rangle \rightarrow |11\rangle.$$

Bob then performs a projective measurement in the computational basis $\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\}$ with operators

$$\{|00\rangle_{BA} \langle 00|, |01\rangle_{BA} \langle 01|, |10\rangle_{BA} \langle 10|, |11\rangle_{BA} \langle 11|\},$$

where

$$|ba\rangle_{BA} = |b\rangle_B \otimes |a\rangle_A,$$

so the first bit corresponds to Bob's qubit and the second to Alice's qubit.

D. Nonlocality of ‘message’

At first we need to refresh our understanding of reduced density operator [4, see page 106]. To describe the state of a subsystem of a composite quantum system, we use the reduced density operator. Suppose a composite system AB is described by a density operator ρ_{AB} . The state of subsystem A is defined as

$$\rho_A = \text{Tr}_B(\rho_{AB}),$$

where Tr_B denotes the partial trace over system B . [5]

The reduced density operator for system A represents the state of system A , in the sense that it reproduces the correct measurement statistics for all measurements performed on A .

$$\text{Tr}(\rho_A M_A) = \text{Tr}(\rho_{AB}(M_A \otimes I_B)).$$

For a detailed justification of this identification, see [4, Box 2.6, page 107].

Take-home message: *the reduced density operator provides all information accessible to an observer with access only to subsystem A .*

Note :

$$\Rightarrow \quad \rho_i = \frac{I}{2} \text{ yields identical measurement statistics for all measurements. } \quad i \in \{A, B\}$$

Physically: the measurement yields no useful information (complete uncertainty).

Our aim here is to show that the encoded message is not locally stored and cannot be extracted locally.

We begin with the Bell state

$$|\Phi^+\rangle_{BA} = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

Its density matrix is

$$\rho_{BA} = |\Phi^+\rangle\langle\Phi^+| = \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|).$$

Tracing out Alice's qubit gives Bob's reduced state:

$$\rho_B = \text{tr}_A(\rho_{BA}) = \frac{1}{2}(\text{tr}_A(|00\rangle\langle 00|) + \text{tr}_A(|00\rangle\langle 11|) + \text{tr}_A(|11\rangle\langle 00|) + \text{tr}_A(|11\rangle\langle 11|)).$$

Using

$$\text{tr}_A(|00\rangle\langle 00|) = |0\rangle\langle 0|, \quad \text{tr}_A(|00\rangle\langle 11|) = 0, \quad \text{tr}_A(|11\rangle\langle 00|) = 0, \quad \text{tr}_A(|11\rangle\langle 11|) = |1\rangle\langle 1|,$$

we obtain

$$\rho_B = \frac{1}{2}(|0\rangle\langle 0| + |1\rangle\langle 1|) = \frac{I}{2}.$$

Thus Bob's reduced state is maximally mixed. Similarly,

$$\rho_A = \text{tr}_B(\rho_{BA}) = \frac{I}{2}.$$

So neither subsystem carries the message locally.

Now let Alice apply a local unitary U to her qubit. The global state becomes

$$\rho'_{BA} = (I \otimes U)\rho_{BA}(I \otimes U^\dagger).$$

Bob's reduced state is then

$$\rho'_B = \text{tr}_A(\rho'_{BA}) = \text{tr}_A[(I \otimes U)\rho_{BA}(I \otimes U^\dagger)] = \text{tr}_A(\rho_{BA}) = \rho_B = \frac{I}{2}.$$

Hence Bob's local state is unchanged by Alice's encoding.

In particular,

$$\rho_B^{(I)} = \rho_B^{(X)} = \rho_B^{(Z)} = \rho_B^{(XZ)} = \frac{I}{2},$$

so all encoded states give the same local state for Bob. Therefore, for any measurement $\{M_i\}$ on Bob's subsystem,

$$p(i) = \text{Tr}(\rho_B M_i) = \frac{1}{2} \text{Tr}(M_i), \quad \forall \{M_i\},$$

which is independent of Alice's choice of encoding.

The conclusion is pretty straightforward. The encoded message is not stored in either qubit separately, and it cannot be extracted from Bob's subsystem alone .

E. Implementation using Qiskit

The superdense coding protocol was implemented using the `Qiskit` framework [6] (version 2.4.1). The circuit was simulated using the `AerSimulator` backend from `Qiskit Aer` [7] (version 0.17.2), and executed on IBM Quantum hardware [8].

For hardware execution, the circuit was transpiled to match the native gate set and connectivity of the selected backend, `ibm_marrakesh`, using `Qiskit`'s transpilation tools. The experiments were carried out using the IBM Quantum Runtime service, with results obtained from 8192 measurement shots.

The superdense coding protocol was implemented in a Google Colab environment using `Qiskit`, following available IBM Quantum documentation and tutorials on SDC, with minor AI assistance for code formatting.

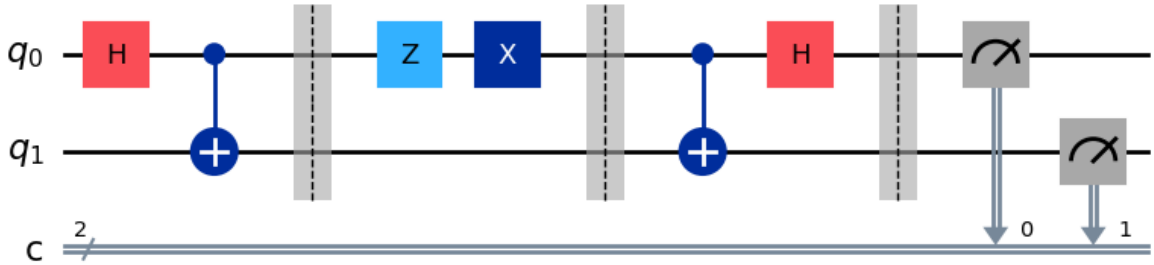


FIG. 3. `Qiskit` implementation of the superdense coding circuit.

1. Simulation on `AerSimulator`

Prior to hardware execution, the circuit is tested using the `AerSimulator`, which enables ideal (noise-free) simulation.

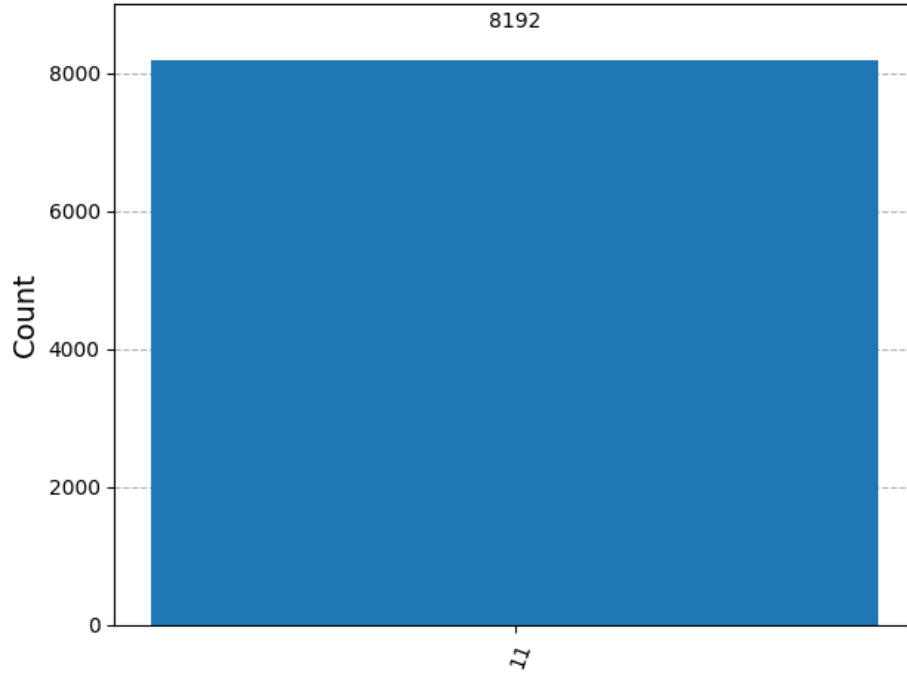


FIG. 4. Ideal simulation of the SDC using `AerSimulator`, showing deterministic recovery of the state 11.

2. Execution on IBM Quantum Hardware

The protocol was then executed on IBM's `ibm_marrakesh` backend. The result in the Fig. 5 shows the correct output 11 remains dominant but small deviations appear due to hardware noise.

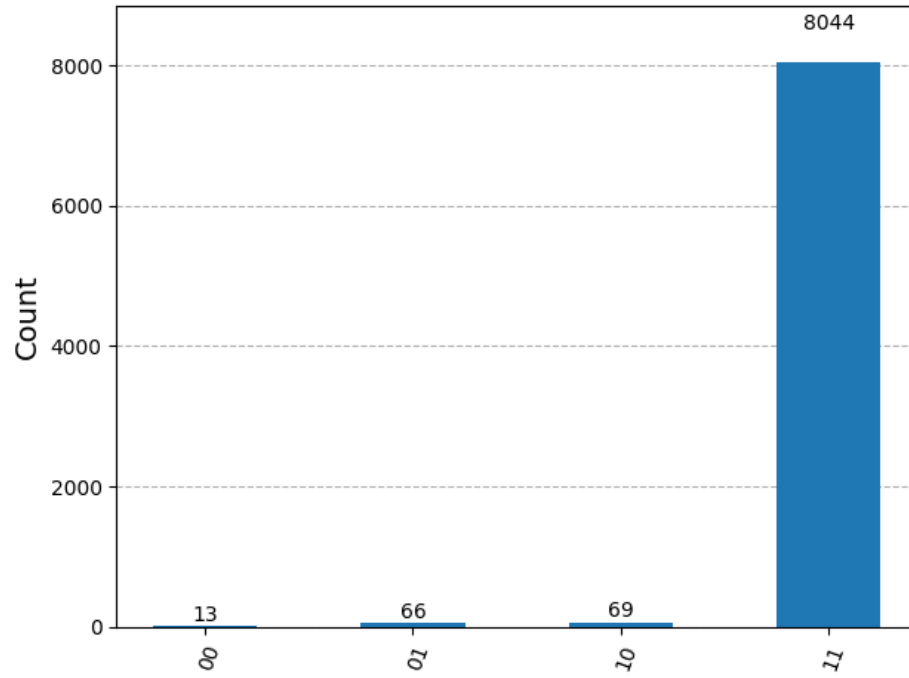


FIG. 5. Results of the superdense coding protocol run on IBM's `ibm_marrakesh` quantum backend. The correct outcome 11 is obtained in 8044 out of 8192 shots, with smaller counts in other outcomes due to hardware noise.

III. QUANTUM TELEPORTATION

a. Introduction to Quantum Teleportation Quantum teleportation is a protocol that allows the transfer of an unknown quantum state from one location to another using shared entanglement and classical communication. In this process, Alice and Bob initially share an entangled pair of qubits. Alice also possesses a qubit in an unknown state that she wants to send to Bob. By performing a joint (Bell basis) measurement on her qubits and sending two classical bits of information to Bob, he can apply an appropriate unitary operation to recover the original quantum state. Importantly, the state is not physically transmitted, but rather its information is transferred, and the original state is destroyed in the process.

A. Protocol

Quantum teleportation is a protocol by which an unknown quantum state can be transferred from Alice to Bob using one shared entangled pair and two bits of classical communication. Suppose Alice wants to teleport the unknown qubit

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle,$$

Alice and Bob share one maximally entangled EPR pair,

$$|\beta_{00}\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

The first qubit belongs to Alice and the second qubit belongs to Bob. Therefore, the complete initial three-qubit state is

$$|\psi_0\rangle = |\psi\rangle|\beta_{00}\rangle.$$

$$\Rightarrow |\psi_0\rangle = (\alpha|0\rangle + \beta|1\rangle) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle).$$

$$\Rightarrow |\psi_0\rangle = \frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|100\rangle + \beta|111\rangle].$$

Now Alice applies a CNOT gate with the first qubit as the control qubit and the second qubit as the target qubit. Therefore, after the CNOT gate, the state becomes

$$|\psi_1\rangle = \frac{1}{\sqrt{2}}[\alpha|000\rangle + \alpha|011\rangle + \beta|110\rangle + \beta|101\rangle].$$

Next, Alice applies the Hadamard gate on the first qubit. So,

$$|\psi_2\rangle = \frac{1}{\sqrt{2}} [\alpha H|0\rangle|00\rangle + \alpha H|0\rangle|11\rangle + \beta H|1\rangle|10\rangle + \beta H|1\rangle|01\rangle].$$

$$\Rightarrow |\psi_2\rangle = \frac{1}{2} [\alpha(|0\rangle + |1\rangle)|00\rangle + \alpha(|0\rangle + |1\rangle)|11\rangle + \beta(|0\rangle - |1\rangle)|10\rangle + \beta(|0\rangle - |1\rangle)|01\rangle].$$

Now we group the state according to Alice's first two qubits:

$$\begin{aligned} |\psi_2\rangle = \frac{1}{2} [& |00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) \\ & + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)]. \end{aligned}$$

Now it is clear that after Alice measures her two qubits, Bob's qubit collapses into one of four possible states. Depending on Alice's measurement result, Bob can apply a suitable quantum gate to recover the original state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

In the following table, depending on Alice's measurement, Bob is going to apply the following gate on his qubit

Alice's measurement	Bob's collapsed state	Gate applied by Bob
00	$\alpha 0\rangle + \beta 1\rangle$	I
01	$\alpha 1\rangle + \beta 0\rangle$	X
10	$\alpha 0\rangle - \beta 1\rangle$	Z
11	$\alpha 1\rangle - \beta 0\rangle$	ZX

In this way the 1st qubit is teleported to Bob.

The circuit shown below represents the quantum teleportation protocol. The first qubit is prepared in an arbitrary unknown state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

and 2nd and 3rd qubit are the bell pair.

B. Circuit Diagram

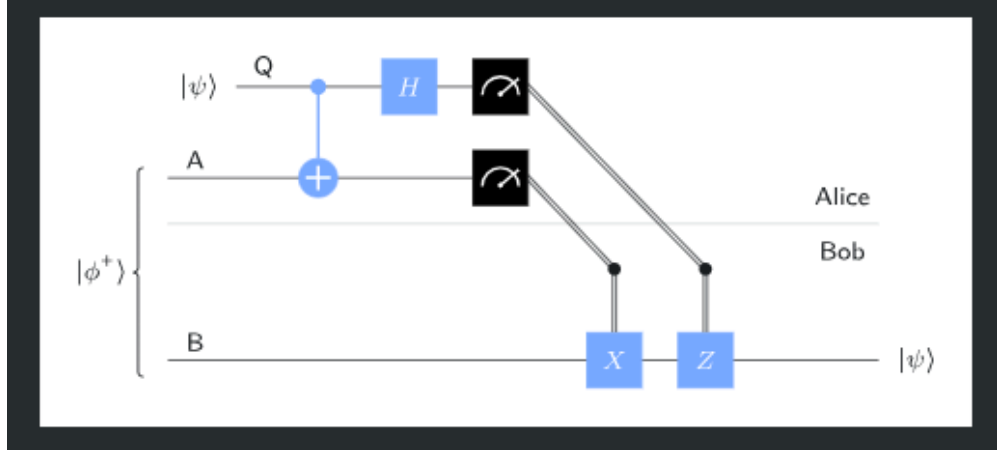


FIG. 6. Quantum Teleportation Circuit

C. Simulation in Qiskit

To verify the quantum teleportation protocol, the circuit was implemented and simulated using Qiskit. The simulation uses three qubits: the first qubit contains the arbitrary unknown state, the second qubit belongs to Alice, and the third qubit belongs to Bob. The arbitrary input state is prepared using the unitary gate

$$U(\theta, \phi, \lambda),$$

where in the simulation we choose

$$\theta = \frac{\pi}{3}, \quad \phi = \frac{\pi}{4}, \quad \lambda = \frac{\pi}{5}.$$

After preparing the unknown state on the first qubit, an entangled Bell pair is created between Alice's second qubit and Bob's third qubit by applying a Hadamard gate followed by a CNOT gate. Alice then apply a CNOT gate between the unknown qubit(Control qubit) and her entangled qubit(target qubit), followed by a Hadamard gate on the unknown qubit. The two Alice qubits are then measured into two classical registers.

Depending on Alice's measurement results, Bob applies the required correction gates.

To verify that teleportation is successful, the inverse of the original unitary gate is applied to Bob's qubit. If the state has been teleported correctly, this inverse operation brings Bob's qubit back to the state $|0\rangle$. Finally, Bob's qubit is measured.

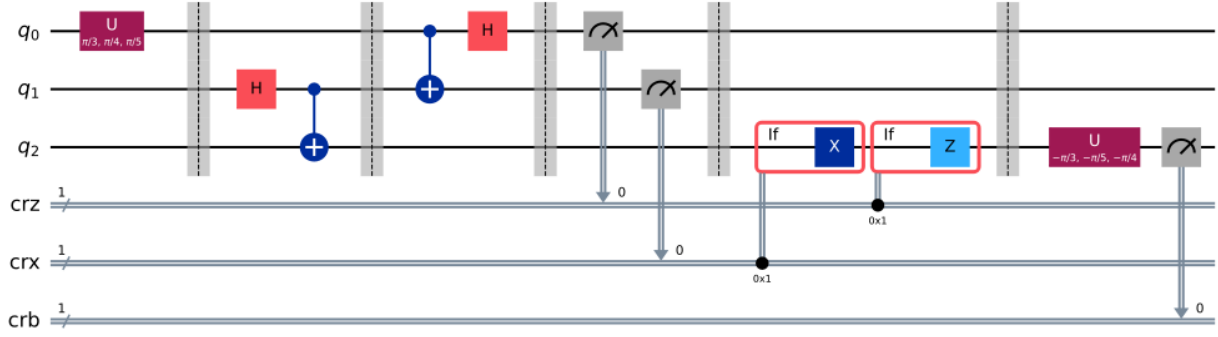


FIG. 7. Quantum teleportation circuit implemented in Qiskit.

The circuit was simulated using the `AerSimulator` backend with 1024 shots. The obtained histogram is shown in Fig. 8. The measurement bit string is ordered as

Bob's bit *crb* Alice's bit *crx* Alice's bit *crz*.

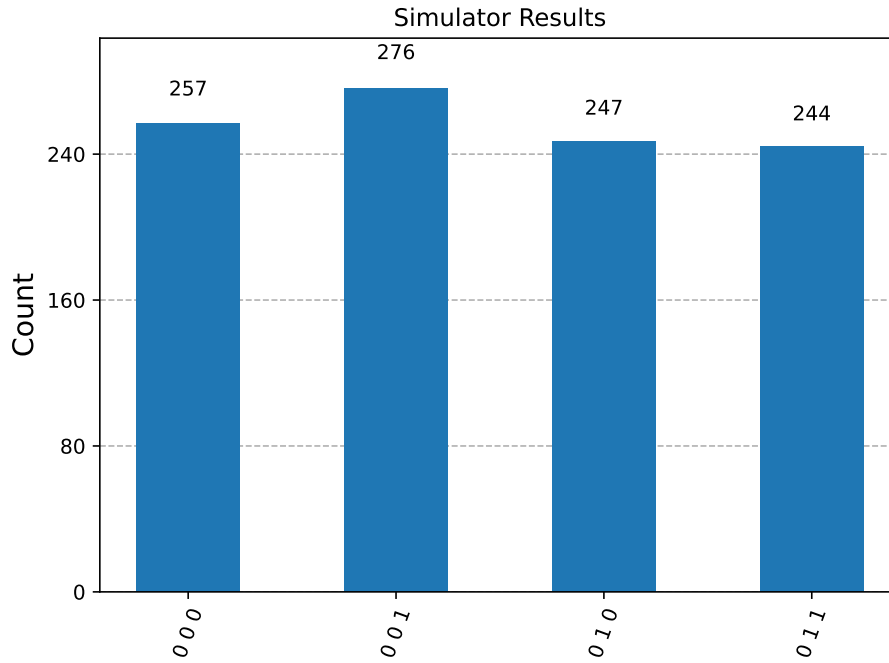


FIG. 8. Simulator result for the quantum teleportation circuit. The leftmost bit, corresponding to Bob's verification measurement, is always zero, confirming successful teleportation

From the histogram, the leftmost bit is always 0, while Alice's two measurement bits are distributed among the four possible outcomes. This shows that after applying the proper correction gates, Bob's qubit is always recovered correctly.

D. Real Hardware Results and Analysis

The teleportation circuit was also executed on real quantum hardware. The measurement outcomes are shown in the histogram in Fig. 9. The bitstring ordering is

(Bob's bit , Alice's bit , Alice's bit).

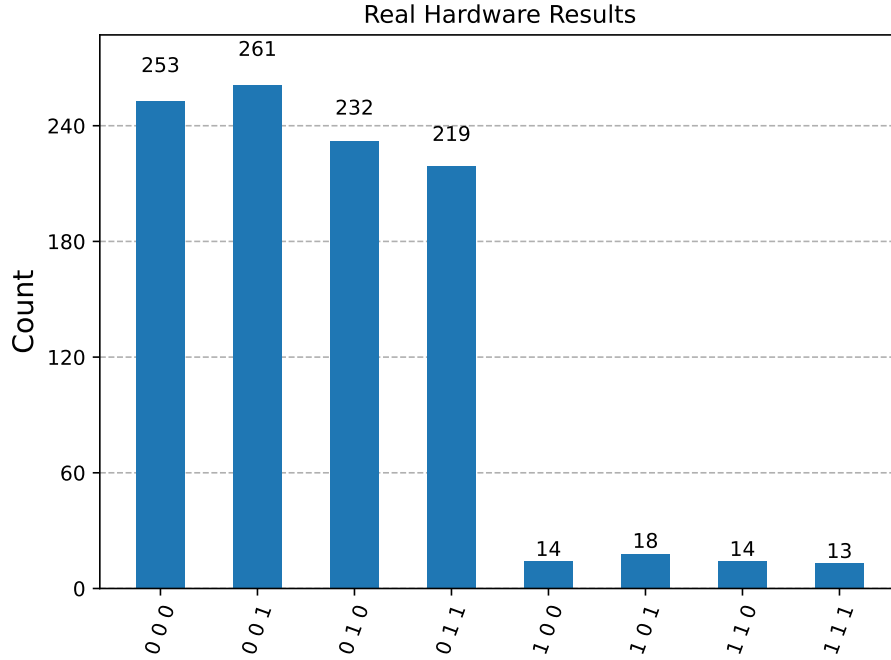


FIG. 9. teleportation in real quantum hardware.

Unlike the ideal simulator case, outcomes where Bob's measurement bit is equal to 1 are now observed. These correspond to the states

100, 101, 110, 111,

which ideally should not appear.

This deviation arises due to noise in real quantum devices, including gate errors, decoherence, and readout errors. These imperfections prevent the correction operations from perfectly reconstructing the original quantum state at Bob's end.

E. Probability of Successful Teleportation

The success of teleportation is determined by the probability that Bob measures 0. From the histogram, the counts are:

$$\text{Bob} = 0: \quad 253 + 261 + 232 + 219 = 965$$

$$\text{Bob} = 1: \quad 14 + 18 + 14 + 13 = 59$$

Thus, the probability of successful teleportation is

$$P_{\text{success}} = \frac{965}{1024} \approx 0.942.$$

This shows that the teleportation protocol works with high fidelity on real hardware, but not perfectly due to noise. The small probability of failure (approximately 5.8%) reflects current limitations of quantum devices.

a. No-Cloning Theorem Quantum teleportation does not violate the no-cloning theorem because it does not copy a quantum state, but transfers it. When Alice performs a joint (Bell basis) measurement, the original state is destroyed due to wavefunction collapse. Bob obtains the state only after applying a correction based on classical information sent by Alice. Hence, two identical copies of the unknown state never exist simultaneously, ensuring consistency with the no-cloning theorem.

b. Where is the Information and Why There is No Faster-Than-Light Communication Initially, the unknown quantum state

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

is completely with Alice. After applying the entangling operations (CNOT and Hadamard), the global three-qubit state becomes

$$|\psi_2\rangle = \frac{1}{2} \left[|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle) \right].$$

At this stage, the information (α, β) is no longer in a single qubit; it is encoded non-locally in the correlations between all three qubits.

After Alice performs her measurement, Bob's qubit becomes one of the four states

$$|\psi\rangle, X|\psi\rangle, Z|\psi\rangle, ZX|\psi\rangle,$$

but he does not know which one. Hence, his reduced density matrix is

$$\rho_B = \frac{1}{4} \left(|\psi\rangle\langle\psi| + X|\psi\rangle\langle\psi|X + Z|\psi\rangle\langle\psi|Z + ZX|\psi\rangle\langle\psi|ZX \right) = \frac{I}{2}.$$

This is a maximally mixed state, so Bob cannot extract any information about $|\psi\rangle$.

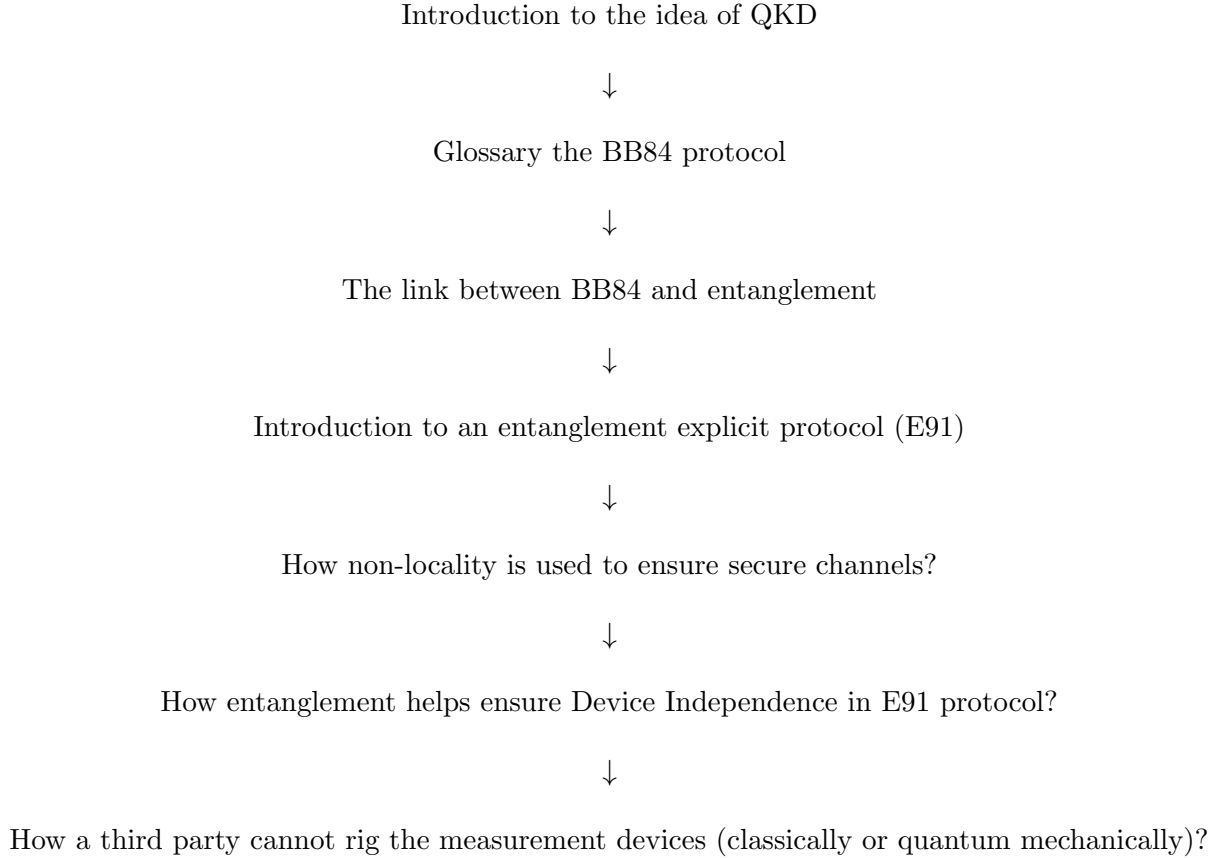
Only after Alice sends two classical bits does Bob know which correction to apply, and then he recovers

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle.$$

Therefore, no information is available to Bob before classical communication, and teleportation does not allow faster-than-light transmission.

IV. QUANTUM KEY DISTRIBUTION AS AN APPLICATION OF ENTANGLEMENT

A. Section Roadmap



B. Introduction and Motivation

We now turn to investigate onto the usage of entanglement as a way to securely transfer top secret information between two parties.

Premise: We go with the very creative names Alice and Bob referring to the two parties. Suppose Alice wants to send Bob some information which she has converted to bits. Now if they mutually could decide on some key only known to them, then Alice could simply add (or do any invertible operation) using those key and Bob could simply invert the operation to retrieve the message without anyone knowing the message. But doing so requires a new key for each message sent, otherwise if the message is ever publicly known, eavesdroppers of Alice Bob channel could decipher/ reverse-engineer the key. Thus a new key each for each message is necessary for privacy.

This idea is at the heart of QKD.

Thus this reduces our aim from securely sending the message to be able to generate key **only** known to the two parties namely Alice and Bob.

Now we turn a blind eye to all classically interesting ways of cryptographic texting your partner and look into quantum ways of doing so. More importantly we'll see how using entanglement might give a upper-hand to cryptographic communication between Alice and Bob.

1. *Why bother Quantum at all?*

Classical Cryptography depends on computational hardness like factoring large numbers (used in RSA) they assume that certain problems are hard to solve, if someone finds a fast enough algorithm, security breaks.

This is where two of our newly established Quantum pillars roll in.

No-cloning Theorem ensures that any eavesdropper just cannot make a copy of the encryption and keep it with itself. Its simply not possible unlike the classical analogue.

Measurement Disturbance i.e. perturbing the system by any means to get any information out of the system disturbs the intended message altogether.

These two would be our principal advantages of QKD over its classical analogue while other advantages might depend case to case and protocol to protocol.

C. BB84 Protocol

BB84 is a prepare and measure quantum key distribution protocol given by Bennett and Brassard in 1984 [1], hence the name. For this section, I have follwed [4] , [9] , [10] The idea is as follows:

Alice prepares a series of qubits in either $|0\rangle$ or $|1\rangle$ in either of **X** or **Z** bases and sends them to Bob. Bob has the independence to measure in any the above bases. When Bob measures a

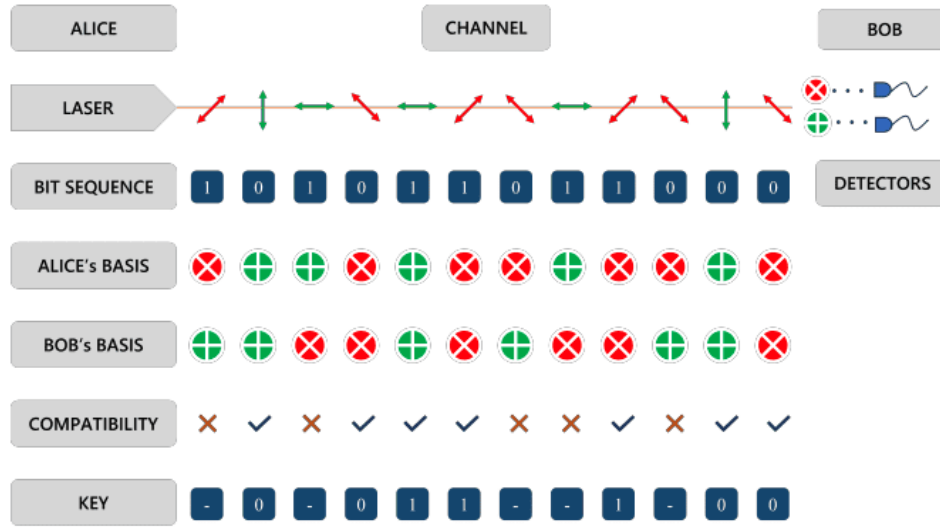


FIG. 10. Schematic representation of the BB84 protocol (adapted from [11]).

qubit in the base in which it is prepared, he gets the same qubit. When measured elsewhere, he gets randomly one of $|0\rangle$ or $|1\rangle$, which of course is not reliable. Both Alice keeps a note of the states prepared with herself and Bob keeps a note of the states measured with himself.

What they do to eliminate the unreliable measurements is by releasing publicly the ordered set of bases used to prepare and bases used to measure. Then both retaining the states for which the bases match. Thus they have their key.

But, we ask, how does this ensure no eavesdropping? This question arises for when both the bases match (which again depends on chance) but as it would turn out Eve (we call the eavesdropper very creatively Eve) had already measured a qubit in the sequence they now have as their key. This makes even the thus formed "reliable" set as unreliable. This problem is not only associated with third party perturbation but even during transmission, we must entertain errors like bit-flip error which doesn't need a third party intervention.

To tackle such problems, out of say N key bits, both Alice and Bob publicly release m (a fraction of N) and check for inconsistencies. We accept our key up to certain percentage of error and reject otherwise, forcing us to do this process again and again until the error is under that bound.

In proper literature this is called Quantum Bit Error Rate or **QBER**

$$QBER = \frac{\text{Number of mismatched bits}}{\text{Total compared bits}}$$

There are more caveats to explore in the context of the BB84 Protocol like error correction but we'll avoid that for that goes quite off our intended course.

1. Entanglement Interpretation of BB84

Although standard BB84 has the description of a prepare and measure protocol but it can equivalently interpreted as an entanglement based protocol.

Instead of Alice preparing a state and sending it to Bob, we now have Alice preparing an entangled pair and measuring one of the qubits and sending the rest to Bob. The correlation between qubits of an entangled pair ensures that this process replicates the results of the prepare and measure interpretation of the protocol.

Prepare and Measure $\equiv$ Measuring one half of the entangled pair

All this is not in vain, there are certain advantages of using an entangled pair over the prepare and measure protocol. This bridges us to another QKD protocol that explicitly uses entanglement, where **entanglement as a resource** will become more apparent!

D. E91 Protocol

You'll be surprised to see how similar this is to BB84.

Introduced by Arthur Ekert in 1991 [12], hence the name. Quantum Cryptography in E91 protocol is based on Bell theorem and entanglement. This protocol. This protocol employs the concept of entanglement to detect any eavesdropping attempts.

1. Build-Up

Quantum Gates:

For the purposes of this protocol, Alice and Bob measure in the bases $\mathbf{X}$, $\mathbf{Z}$ and normalized linear combinations of them.

$$\mathbf{W} = \frac{\mathbf{X} + \mathbf{Z}}{\sqrt{2}} \tag{9}$$

$$\mathbf{V} = \frac{-\mathbf{X} + \mathbf{Z}}{\sqrt{2}} \tag{10}$$

CHSH Inequality:

As already explained in **Section 1.6**,

$$\begin{aligned}\langle S|S\rangle &= E(a_1, b_1) + E(a_2, b_1) + E(a_1, b_2) - E(a_2, b_2) \\ E(a_i, b_j) &= \langle a_i * b_j | a_i * b_j \rangle\end{aligned}\tag{11}$$

Where a_i, b_j are measurement bases of Alice and Bob respectively.

Since maximally entangled bell states corresponds to the $2\sqrt{2}$ violation of CHSH Inequality , therefore benefits to the security against any eavesdropping which causes the correlation $\langle S|S\rangle$ below this limit. Thus, the CHSH inequality acts as an additional safeguard, ensuring the security of keys production.

2. Protocol

In this section which, we provide a summary on how the E91 protocol works. Bases in this protocol refer to the same quantum gates above. These gates are used for bases measurement. Steps for constructing a simple E91 protocol are as follows[13]:

- The E91 protocol uses entangled states. In this paper, we use singlet state of

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.\tag{12}$$

- A trustful source generates pairs of $\frac{1}{2}$ -spin particles in a singlet or an entangled state and sends them to Alice and Bob.
- Of each particle, Alice and Bob measure an observable, chosen randomly and independently from each other from a set of complementary attributes.
- Alice measures with angles $\theta_A^i = \{0, \frac{\pi}{4}, \frac{\pi}{2}\}$ or similar to $\{X, W, Z\}$, while Bob measures with $\theta_B^j = \{\frac{\pi}{4}, \frac{\pi}{2}, \frac{3\pi}{4}\}$ or similar to $\{W, Z, V\}$.

Why 3 bases each and not 2?

These help generate keys and measure CHSH inequality simultaneously. A set of 2 pairs decided to be used for CHSH inequality and the remaining pair is used for key generation.

We want the key pair to have perfect correlation thus good choices would be $(A_3, B_2)\{\text{both Z}\}$ or $(A_2, B_1)\{\text{both W}\}$.[\[14\]](#)

- Each random measurements can result in a spin state of $|1\rangle$ (spin up) or $|0\rangle$ (spin down), whereby they show a correlation coefficient E of a certain observable to have a certain value calculated by

$$E(a_i, b_j) = P_{++}(a_i, b_j) + P_{--}(a_i, b_j) - P_{+-}(a_i, b_j) - P_{-+}(a_i, b_j). \quad (13)$$

- After that, they exchange the bases of their measurement over the public channel and divide the results into two groups:

- one with according bases
- one with discarding bases

- The ones with according bases determine the key as they are perfectly correlated (which means they have the exact value) due to their entangled state expressed as

$$E(a_2, b_1) = E(a_3, b_2) = 1. \quad (14)$$

- The discarding particles can still be used to test either Eve listens to the qubits on the quantum channel or not.
- The Bell inequality is able to test the presence of an eavesdropper.
- Alice and Bob can now exchange the discarding particles openly (over the public channel), since they do not participate in the composition of the key.
- These values determine whether or not $\langle S|S \rangle$ is fulfilled:
 - if so, an entangled state is present and the key derived from the group of corresponding result is legitimate
 - if not, Eve may have altered some or all of the particles and the key is not secure

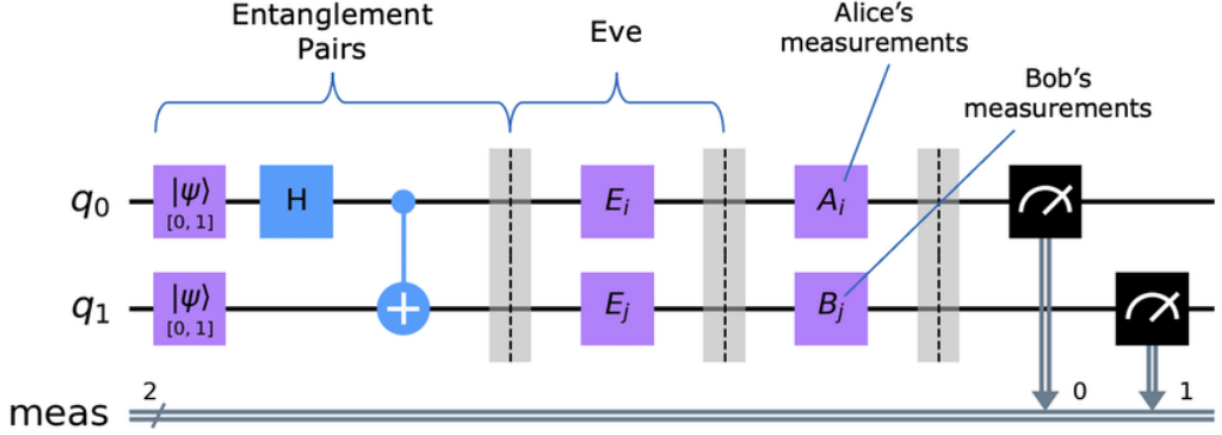


FIG. 11. Qiskit Circuit Diagram of E91 Protocol(adapted from [15])

3. Key Considerations

Security :

The juice of the entire argument that is going to follow is this:

$$\text{CHSH Inequality: } |S| > 2 \implies \text{non-local quantum correlations} \quad (15)$$

Recall that we have three sets that partition the entirety of measurements done. Key set has the measurements when both chose the respective base from key pair , test set had the keys the measurement with same bases but from the test pairs , and third set containing all the measurements done in different bases. Both the parties, Alice and Bob are made to release the test set publicly and $\langle S|S \rangle$ as defined in the CHSH Inequality. Since we are using maximally entangled bell state, we must get $2\sqrt{2}$ for the case if there were no external perturbation or Quantum Channel Noise. So any deviation from this limit would relay imperfections in the system.

Now that we notice even transmission errors also cause this deviation, we choose a certain threshold , if a test set goes below that, we consider the system compromised and redo the whole process.

Device Dependency :

In BB84 one assumes that state preparation and measurement devices are perfect. Even these error are taken into account by the virtue of entanglement in E91 protocol and in this context E91 turns out to be superior to BB84.

In E91 protocol, one does not need to trust any device at all, all the in-discrepancies are detected via the CHSH Inequality. For E91, one only assumes

- Alice and Bob choose the measurement independently
- No communication between the devices during measurement i.e. they are space-like separated

What if Eve rigs the devices that Alice and Bob has ?

Eve could have

- preprogrammed outputs
- embedded hidden variables i.e.

there is a variable λ not accessible to either of Alice or Bob. Thus outputs that they get is now also a function of λ too. Thus once λ is fixed, outputs are predetermined.

Nothing goes against our assumptions, so we must be able to detect if something like this is done. All of this is classical and CHSH Inequality forbids any classical system to produce correlation $\langle S|S \rangle$ over 2.

Each device produces outputs based on its own inputs and some hidden variables. To produce non-local correlations, devices cannot be locally programmed.

What if Eve finds a classical loophole?

She could read do measurements on the qubits (which reduces $\langle S|S \rangle$ and make the devices internally) communicate so that $\langle S|S \rangle$ increases. This is something that is forbidden to do , it is made sure that the devices could not communicate and are space-like separated, and thus such a strategy would never work.

Can Eve only rig the device classically?

Interestingly no, Eve is free to come up with any strategy she wants thus she could allegedly "Quantum Rig" the system too. So we should be able to detect that right? Yup, by establishing non-locality with the two devices, she correlates herself with the two parties. Now, there is something called Monogamy of Entanglement which helps us decode this case better.

Monogamy of Entanglement ensures that two strongly correlated systems cannot have a strong correlation with a third party.

Proof: The monogamy of entanglement can be formalized through the Coffman–Kundu–Wootters inequality [16], which for a tripartite system (A,B,E) states that

$$C_{AB}^2 + C_{AE}^2 \leq C_{A(BE)}^2. \quad (16)$$

For a pure state, $C_{A(BE)} = 1$. If Alice and Bob share maximal entanglement $C_{AB} = 1$, it follows that $C_{AE} = 0$, implying that no entanglement can be shared with a third party. This establishes the monogamous nature of quantum entanglement.

Thus again this rigging by Eve will also be caught in $\langle S|S \rangle$. If not then she has too little information about the system.

Thus by no means will Eve be able to gain *significant* information of the system or try to manipulate the results without the malpractices being detected.

Device Independent QKD in general requires space-like separation and no communication between devices.

E. Summary

Our aim to bring in this section was to explore possibilities of using entanglement as a way to secure data transfer. Here are the key take-home messages.

- The Quantum Bit Error Rate (QBER) serves as an operational measure of channel security and eavesdropping.
- Violation of the CHSH inequality certifies the presence of non-local correlations inaccessible to classical strategies.
- The E91 protocol leverages entanglement explicitly, enabling security verification through Bell inequality violations.
- Security in E91 arises from entanglement via Bell non-locality and monogamy.
- Security in E91 does not rely on trusting the internal workings of measurement devices.
- Entanglement acts as a physical resource that constrains information sharing and enables secure key distribution.

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$$\text{Tr}_B(|a_1\rangle\langle a_2| \otimes |b_1\rangle\langle b_2|) = |a_1\rangle\langle a_2| \text{Tr}(|b_1\rangle\langle b_2|),$$

where $|a_1\rangle, |a_2\rangle$ are vectors in the state space of A , and $|b_1\rangle, |b_2\rangle$ are vectors in the state space of B . The trace on the right-hand side is the usual trace over system B , so that

$$\text{Tr}(|b_1\rangle\langle b_2|) = \langle b_2|b_1\rangle.$$

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